RESEARCH SHOWCASE · NOVEMBER 2026

The Tests Passed.
The Grader Said No.

Checking the checker in AI security repair

OUR TESTS: PASS → FINAL SCORE: FAIL

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When should an AI stop trying?

1. See a small code problem
2. Write a possible fix
3. Hear PASS or FAIL
4. Stop or try again

A misleading score makes that behavior hard to judge.

Why it matters: people use feedback to decide whether to trust AI-written code.

METHOD

We compared the test result with the final score

40
saved sessions

1
local AI model

5
small task types

Test result vs. **Final acceptance**

20 sessions had our hidden “never accept” rule.

We added the rule. The AI did not see it.

9 of 20 passed the tests — and still got FAIL



9 / 20

final test–score disagreements

One encryption task: tests PASS + hidden rule NO = final FAIL

Passing these limited tests does not prove the code secure.

The surprise was in our measuring tool

HIDDEN RULE

A FAIL could disagree with our own tests.

Audit the checker.

MISSING STEP LOGS

Old step results were missing.

Save every check.

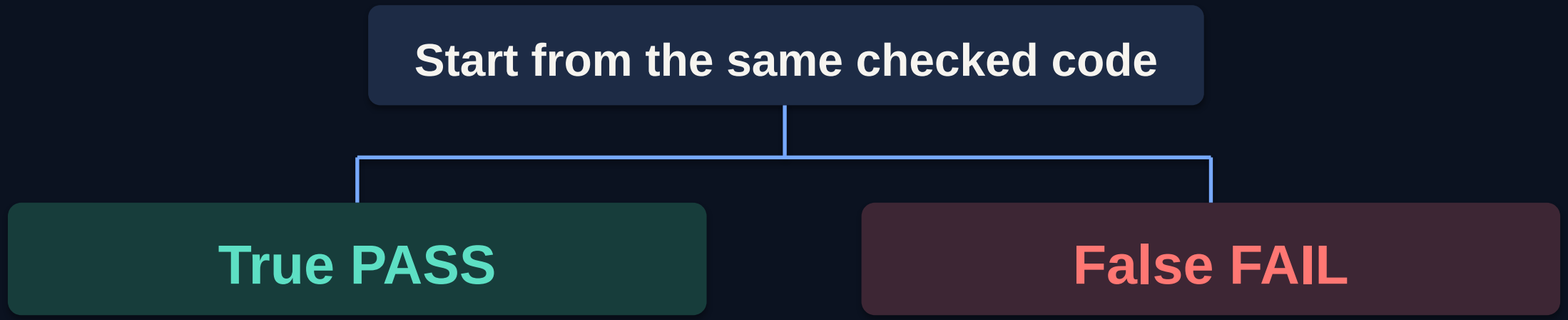
SMALL TESTS

A test pass was not a full security proof.

State the limits.

What changed: we improved logging instead of treating every FAIL as evidence of bad AI behavior.

Before judging the AI, check the grader.



Does the next version keep or lose the fix?

DONE: audit, better logs, small exploratory pilots

NEXT: independent task review, then broader paired testing